CB CareerBridge AI ENTERPRISE EDITION

Complete Architecture, Design, & UI Specification

AI-Powered Career Intelligence, Educational Guidance, and Full-Cycle Recruitment Ecosystem

"Bridge Your Skills. Build Your Tomorrow."

FastAPI v1 Architecture | React 19 + TypeScript | Three.js 3D WebGL Canvas | Deterministic 70/30 Match Engine | SQLAlchemy 2.0 ORM | 15 Relational Entities | 100% Offline AI Intelligence | 17/17 Automated Tests Passing

COURSE / PROJECT CONTEXT
AI Assisted Coding — Final Capstone Project

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TARGET ARCHITECTURE
Full-Stack Enterprise Web Application

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1. Executive Summary & Project Vision

1.1 The Problem Space

In the modern technical recruitment landscape, both students and recruiters face severe systemic disconnects: - **Students** lack actionable clarity regarding the exact skills demanded by industry roles. Traditional job descriptions are verbose, vague, and uncalibrated, leaving candidates unsure whether their academic preparation qualifies them for entry-level or junior engineering positions. - **Recruitment Platforms** rely on black-box keyword matching algorithms that penalize qualified candidates due to arbitrary keyword phrasing or provide candidates with zero transparent feedback when an application is rejected. - **Skill Acquisition** is fragmented. Students study disparate courses without an integrated roadmap that connects self-study directly to verified job openings and interactive mock interview evaluation.

1.2 The CareerBridge Solution

CareerBridge AI is an enterprise-grade, full-stack career intelligence, educational guidance, and transparent recruitment platform designed to eliminate the ambiguity between student capability and industry demand.

CAREERBRIDGE AI ECOSYSTEM	
STUDENT CAPABILITIES	RECRUITER CAPABILITIES
• Verified & Self-Reported Skill Competencies • Deterministic Skill-Gap Radar Evaluation (6 Axes) • Dynamic Milestone Roadmaps • ATS Resume Diagnostics • AI Mock Interview Rubrics • Explainable 70/30 Job Match	• Company Profile & Brand Showcase • Multi-Criteria Job Posting Engine • Transparent Candidate Pipeline • Application Status Transitions (Submitted → Shortlist → Offered) • Skill-Based Talent Search • Market Skill Demand Intelligence • Privacy-Guarded Direct Outreach

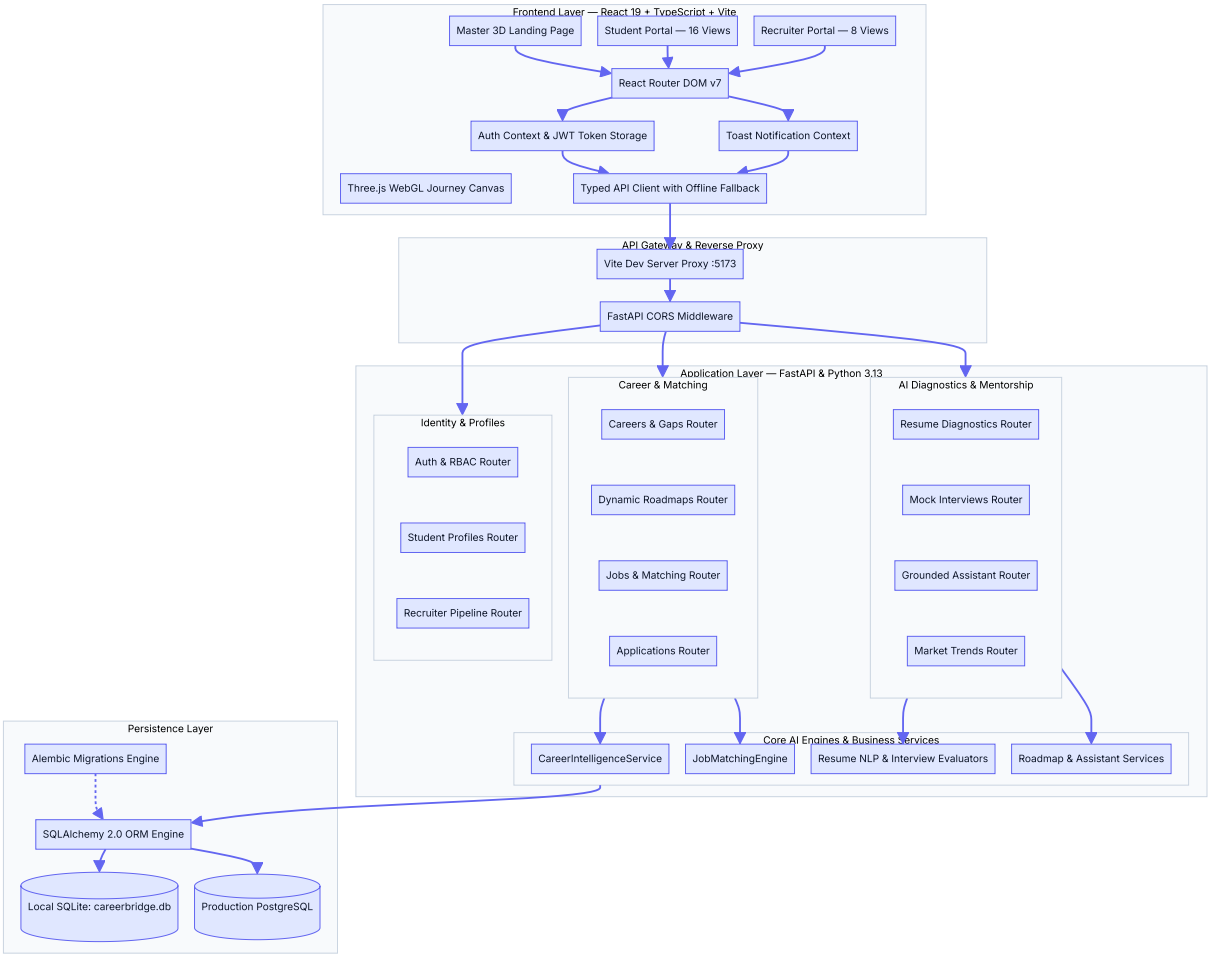
1.3 Key Innovations

1. **Explainable Deterministic Matching**: Unlike opaque scoring models, CareerBridge utilizes a deterministic matching engine that breaks down candidate-job alignment into **70% Required Skills**, **30% Preferred Skills**, and a **+5% Project Evidence Bonus**.
2. **Local Intelligent AI Architecture**: The platform operates with 100% offline intelligence out-of-the-box (zero mandatory external API dependencies) via deterministic mathematical algorithms, heuristic NLP, and curated question rubrics, while maintaining plug-and-play compatibility with Gemini, OpenAI, and Anthropic.
3. **Immersive 3D WebGL Storytelling**: A custom Three.js continuous spline journey canvas seamlessly guides first-time visitors from initial skill exploration down to career triumph with smooth scroll-coupled camera interpolation.
4. **Cinematic Dark Glassmorphism UI**: High-contrast, accessibility-tested visual tokens (`#05070D` background, frosted glass panels with 24px backdrop blur, vibrant lavender/cyan accents) deliver an exceptionally modern desktop experience.

2. High-Level System Architecture

2.1 System Architecture Diagram

HIGH-LEVEL SYSTEM ARCHITECTURE DIAGRAM — 4-TIER FULL-STACK DECOMPOSITION



2.2 Tier Decomposition & Data Flow

Client Tier

Built with modern web technologies: - **React 19.2.8** featuring concurrent rendering primitives. - **TypeScript 5.8** guaranteeing compile-time type safety across all domain entities. - **Vite 8.3** delivering instant hot module replacement (HMR) and sub-second builds. - **Three.js 0.174** managing the background 3D canvas with custom geometries, ambient particle fields, and mathematical spline curves.

API Gateway & Communication Protocol

- Frontend and backend communicate via **asynchronous RESTful JSON APIs** prefixed with `/api/v1`.
- In development, Vite's internal proxy maps browser requests from `http://localhost:5173/api` directly to `http://127.0.0.1:8000`, resolving cross-origin credential issues and Windows IPv6/IPv4 address resolution mismatches.
- All secure endpoints require the `Authorization: Bearer <JWT>` header.

Application Tier

Powered by **Python 3.13** and **FastAPI**: - Asynchronous route handlers (`async def` and synchronous `def` where database I/O is offloaded to worker threads). - **Pydantic v2** validation models with `ConfigDict(from_attributes=True)` validating inbound JSON bodies and outbound serialization. - Strict Role-Based Access Control (`UserRole.STUDENT` , `UserRole.RECRUITER` , `UserRole.ADMIN`).

Persistence Tier

- **SQLAlchemy 2.0 ORM** employing modern declarative base syntax (`Mapped[T]` and `mapped_column()`).
- Session management through a scoped generator `get_db()` with automatic commit/rollback and connection pooling.

2.3 Dual-Database Architecture (SQLite & PostgreSQL)

To enable frictionless academic evaluation and seamless enterprise deployment: 1. **Local Zero-Setup Mode (Default)**: Utilizes SQLite (`sqlite:///./careerbridge.db`). Alembic migrations operate with `render_as_batch=True` to support SQLite schema alterations without dropping tables. 2. **Production Mode**: Switching requires only setting `DATABASE_URL=postgresql+psycopg2://user:pass@host:5432/careerbridge` in `.env` . The ORM models and Alembic migration scripts are 100% dialect-agnostic.

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3. Algorithmic & AI Engine Architecture

3.1 Deterministic Skill Gap & Career Intelligence Engine

Located in `backend/app/services/career_intelligence.py`.

Core Mathematical Formula

The alignment score between a student profile and a target career role is computed deterministically:

$$\text{Alignment Score} = \left(\frac{\sum_{i=1}^{N} \min(S_i, R_i) \cdot W_i \cdot V_i}{\sum_{i=1}^{N} R_i \cdot W_i} \right) \times 100$$

Where: - $S_i \in [1, 5]$: Student's demonstrated proficiency in skill i (defaults to 0 if unacquired). - $R_i \in [1, 5]$: Required proficiency level specified for the target role. - $W_i \in [0.8, 1.5]$: Importance weight of skill i (Critical skills = $1.2 - 1.5$, Recommended = $0.8 - 1.0$). - V_i: Verification factor: - $V_i = 1.00$ for verified skills (demonstrated via linked GitHub project or completed platform assessment). - $V_i = 0.85$ for unverified self-reported skills. - $V_i = 0.70$ for unassessed baseline competencies.

6-Axis Radar Metric Distribution

The engine projects the student's competencies onto six standard industry dimensions: 1. **Algorithms & Computational Foundations** 2. **Systems Architecture & Low-Level Design** 3. **Machine Learning & Statistical Modeling** 4. **Data Engineering & Relational Pipelines** 5. **Software Engineering & Component Architecture** 6. **DevOps, Cloud & Production Infrastructure**

Each axis computes the ratio of student capability against role expectations, returning `current`, `required`, and `cohort_average` values for the interactive radar visualization.

Skill Gap Classification

- **Optimal / Proficient** ($\text{Gap} \leq 10\%$): Marked as verified student strength.
- **Moderate / Needs Improvement** ($10\% < \text{Gap} \leq 30\%$): Targeted for milestone practice.
- **Critical Gap** ($\text{Gap} > 30\%$): Prioritized at the top of the student's dynamic learning roadmap.

3.2 Transparent Job Matching Engine (70/30 Rule + Project Bonus)

Located in `backend/app/services/matching_engine.py`.

Unlike opaque black-box recruiters, CareerBridge implements a mathematically transparent matching model:

TRANSPARENT MATCH FORMULA	
Component	Maximum Weight
Required Skills Match Preferred Skills Match Project Evidence Bonus	70.0% 30.0% +5.0%
TOTAL MAXIMUM MATCH SCORE	100.0% (Clamped)

Algorithmic Procedure

1. **String Normalization**: Skill names are normalized (lowercased, hyphens and slashes replaced with spaces) to ensure resilient matching across naming conventions (e.g., `react.js` matches `React`, `ci/cd` matches `CI-CD`).
2. **Competency Ingestion**: The engine compiles all student skills and scans `student_projects` for demonstrated technologies and tools.
3. **Required Skills Ratio**:

$$\text{Req Score} = \left(\frac{\text{Matched Required Skills}}{\max(1, \text{Total Required Skills})} \right) \times 70.0$$

4. **Preferred Skills Ratio**:

$$\text{Pref Score} = \left(\frac{\text{Matched Preferred Skills}}{\max(1, \text{Total Preferred Skills})} \right) \times 30.0$$

5. **Project Evidence Bonus**: If the student has at least one repository or live project link matching any required skill, an additional $+5.0\%$ is added.
6. **Explainable Breakdown**: The engine returns explicit lists of `matched_skills`, `missing_critical_skills`, and `recommended_to_learn` directly to the client interface.

3.3 ATS Resume NLP Diagnostic Engine

Located in `backend/app/services/resume_service.py`.

Pipeline Architecture

1. **Safe Byte Ingestion**: Accepts PDF or TXT uploads. PyPDF iterates through document pages, extracting plaintext while handling corrupted byte buffers via fallback UTF-8 decoding.
2. **Lexical Skill Extraction**: Matches document text against the platform's relational skill catalogue using regular expression word boundaries (`\b<skill>\b`).
3. **Target Role Comparative Diagnostics**: Compares identified skills against the requested target role's critical requirements, calculating a **Resume Readiness Score (0–100)**: - Found Skills Ratio: up to 50 points - Absence of Missing Critical Requirements: up to 30 points - Structural Quality & Impact Action Verbs: up to 20 points
4. **Action-Verb Enhancement Suggestions**: Scans candidate bullet points for weak verbs (e.g., *"helped with"*, *"worked on"*, *"handled"*) and suggests high-impact alternatives (e.g., *"Architected"*, *"Spearheaded"*, *"Optimized"*, *"Quantified"*).

3.4 Rubric-Grounded AI Mock Interview Simulator

Located in `backend/app/services/interview_service.py`.

Question Bank & Rubric Design

The simulator provides structured questions across four technical specializations: 1. **AI / Machine Learning Engineer** (Self-attention vs cross-attention, gradient explosion, loss divergence diagnostics). 2. **Full-Stack Systems Engineer** (ACID transactions vs eventual consistency, race condition prevention, React 19 server components). 3. **Data Scientist & Quantitative Analyst** (A/B test sample size formulas, feature collinearity,

ROC-AUC calibration). 4. **Cloud & Distributed Systems Architect** (CAP theorem trade-offs, Raft consensus elections, container security isolation).

4-Dimensional Evaluation Rubric

Student responses are graded across four standardized criteria: 1. **Technical Accuracy (0–100)**: Correctness of core architectural assertions and terminology. 2. **Reasoning & Depth (0–100)**: Ability to substantiate claims with first-principles reasoning. 3. **Practical Trade-offs (0–100)**: Awareness of computational cost, latency, scalability, and maintainability. 4. **Clarity & Communication (0–100)**: Conciseness, structure, and professional delivery.

The overall session score is recorded in `interview_sessions` alongside detailed feedback and model benchmark answers.

3.5 Contextual Career Assistant Engine (Zero-Hallucination)

Located in `backend/app/services/assistant_service.py`.

Grounded Knowledge Base

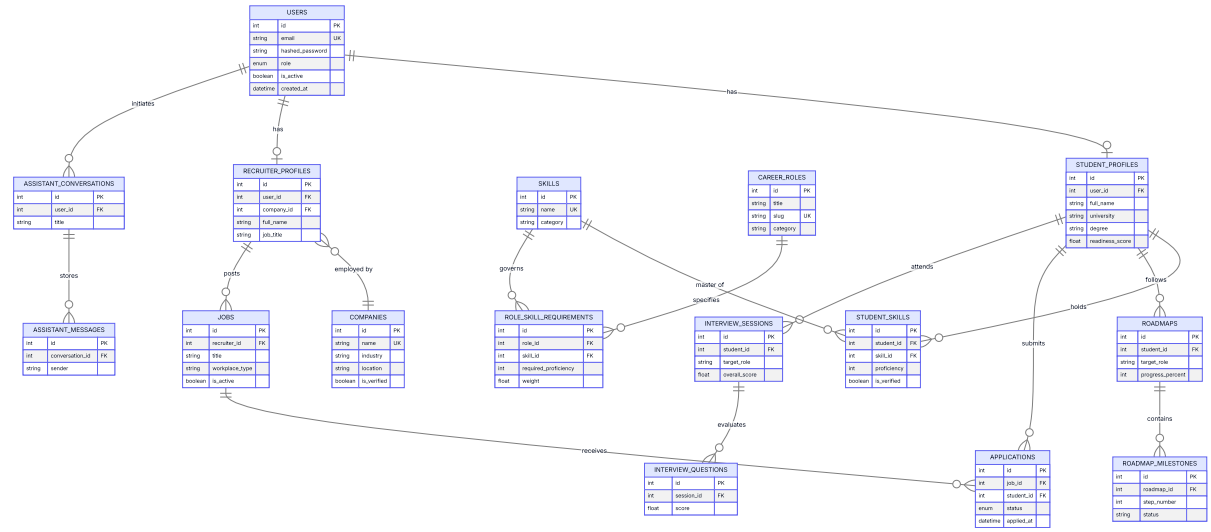
To prevent AI hallucinations, the assistant is deeply integrated with the platform's active database: - Reads the logged-in student's target role, verified skills, and roadmap progress. - Queries active job listings posted by verified employers. - When answering career queries, references real open positions, actual required competencies, and specific platform learning resources. - If an external LLM (Gemini, OpenAI, Anthropic) is configured via `backend/.env`, prompts are injected with grounded JSON context. If the provider times out or fails, the engine falls back to local intelligent generation seamlessly.

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4. Relational Data Models & Database Schema

4.1 Entity Relationship Diagram

ENTITY RELATIONSHIP DIAGRAM — 15 RELATIONAL TABLES & SCHEMA CARDINALITY



4.2 Complete Schema Specifications (15 Relational Tables)

1. `users` Table

Stores primary identity and authentication credentials.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique user identity
`email`	String(255)	Unique, Not Null, Indexed	Login email address
`hashed_password`	String(255)	Not Null	Salted Bcrypt password hash
`role`	Enum	Not Null, Default: `STUDENT`	`STUDENT`, `RECRUITER`, `ADMIN`
`is_active`	Boolean	Default: `True`	Account activation state
`created_at`	DateTime	Default: `utcnow()`	Account creation timestamp
`updated_at`	DateTime	OnUpdate: `utcnow()`	Last profile update timestamp

2. `student_profiles` Table

Stores extended academic and career readiness information.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique student profile ID

Column	Type	Constraints	Description
`user_id`	Integer	FK(`users.id`), Unique, Not Null	Parent user account
`full_name`	String(128)	Not Null	Candidate display name
`avatar_url`	String(512)	Nullable	Profile image URI
`university`	String(255)	Nullable	University or institution
`degree`	String(128)	Nullable	e.g., Bachelor of Technology
`branch`	String(128)	Nullable	e.g., Computer Science & AI
`academic_year`	String(64)	Nullable	e.g., 3rd Year / Junior
`graduation_year`	Integer	Nullable	Projected graduation year
`location`	String(128)	Nullable	Candidate city / region
`target_role`	String(128)	Nullable	Primary targeted career track
`bio`	Text	Nullable	Professional summary
`github_url`	String(255)	Nullable	GitHub profile link
`portfolio_url`	String(255)	Nullable	Personal portfolio URL
`linkedin_url`	String(255)	Nullable	LinkedIn profile link
`career_readiness_score`	Float	Default: `0.0`	Cached aggregate readiness score

3. `recruiter_profiles` Table

Stores recruiter organizational association and contact info.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique recruiter profile ID
`user_id`	Integer	FK(`users.id`), Unique, Not Null	Parent user account
`company_id`	Integer	FK(`companies.id`), Nullable	Associated employer organization
`full_name`	String(128)	Not Null	Recruiter display name
`job_title`	String(128)	Nullable	e.g., Senior Technical Recruiter
`department`	String(128)	Nullable	e.g., Engineering Talent Acquisition
`phone_number`	String(32)	Nullable	Work contact telephone

4. `companies` Table

Represents employer organizations posting roles on CareerBridge.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique company ID
`name`	String(255)	Unique, Not Null	Legal company name

Column	Type	Constraints	Description
`logo_url`	String(512)	Nullable	Company brand logo URI
`website_url`	String(255)	Nullable	Official company URL
`industry`	String(128)	Nullable	Industry sector (e.g., AI Research)
`company_size`	String(64)	Nullable	e.g., 500-1,000 Employees
`location`	String(128)	Nullable	Headquarters location
`description`	Text	Nullable	Overview of company mission
`is_verified`	Boolean	Default: `False`	Platform verification badge

5. `skills` Table

Master taxonomy of engineering and computational skills.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique skill ID
`name`	String(128)	Unique, Not Null	Canonical skill name
`category`	String(64)	Not Null	e.g., Deep Learning, Web Systems
`description`	Text	Nullable	Mathematical & practical definition

6. `student_skills` Table

Association table linking students to acquired skills with evidence.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique student-skill record
`student_id`	Integer	FK(`student_profiles.id`), Not Null	Target student profile
`skill_id`	Integer	FK(`skills.id`), Not Null	Associated master skill
`proficiency`	Integer	Not Null, Default: `1`	Level scale from 1 (Novice) to 5 (Master)
`is_verified`	Boolean	Default: `False`	Verified through project or test
`source`	String(64)	Default: `"self_reported"`	Evidence source (project/assessment)
`evidence_url`	String(512)	Nullable	Direct GitHub repo or commit link

7. `career_roles` Table

Curated industry target roles.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique career role ID
`title`	String(128)	Not Null	e.g., AI / Machine Learning Engineer

Column	Type	Constraints	Description
`slug`	String(64)	Unique, Not Null	URL slug (e.g., `ai-ml`)
`category`	String(64)	Not Null	Industry sector
`description`	Text	Nullable	Role overview
`growth_outlook`	String(64)	Nullable	e.g., +34% YoY Hiring Volume
`avg_salary_range`	String(64)	Nullable	e.g., $125,000–185,000

8. `role_skill_requirements` Table

Skill prerequisites demanded by specific career roles.

Column	Type	Constraints	Description
`id`	Integer	PK, Auto-increment	Unique requirement record
`role_id`	Integer	FK(`career_roles.id`), Not Null	Target role
`skill_id`	Integer	FK(`skills.id`), Not Null	Required skill
`required_proficiency`	Integer	Not Null, Default: `3`	Minimum expected level (1-5)
`importance`	String(32)	Default: `"critical"`	`"critical"` or `"recommended"`
`weight`	Float	Default: `1.0`	Weight multiplier in scoring formula

9. `roadmaps` & `roadmap_milestones` Tables

Structured learning paths and granular milestones. - **`roadmaps`**: `id`, `student_id`, `target_role`, `title`, `description`, `total_weeks`, `progress_percent`. - **`roadmap_milestones`**: `id`, `roadmap_id`, `step_number`, `title`, `description`, `deliverable`, `duration`, `status` (`pending`, `active`, `completed`), `skills_acquired` (JSON list), `resources` (JSON list).

10. `jobs` & `saved_jobs` Tables

Employer openings and student bookmarks. - **`jobs`**: `id`, `recruiter_id`, `title`, `company`, `location`, `workplace_type` (`Remote`, `Hybrid`, `Onsite`), `employment_type` (`Full-time`, `Internship`, `Contract`), `salary_range`, `description`, `category`, `required_skills` (JSON list), `preferred_skills` (JSON list), `is_active`, `created_at`, `deadline`. - **`saved_jobs`**: `id`, `student_id`, `job_id`, `created_at`.

11. `applications` Table

Tracks real candidate job application lifecycles. - Columns: `id`, `job_id`, `student_id`, `status` (`SUBMITTED`, `UNDER_REVIEW`, `SHORTLISTED`, `INTERVIEW`, `OFFERED`, `REJECTED`), `cover_letter`, `applied_at`, `updated_at`.

12. `interview_sessions` & `interview_questions` Tables

Mock interview history and evaluation. - **`interview_sessions`**: `id`, `student_id`, `target_role`, `category`, `difficulty`, `status`, `overall_score`, `created_at`. - **`interview_questions`**: `id`, `session_id`, `question_text`, `student_answer`, `score`, `feedback`, `rubric_scores` (JSON breakdown across 4 criteria).

13. `resume_analyses` Table

Stores parsed resume metrics, ATS keyword matches, and verb suggestions.

14. `assistant_conversations` & `assistant_messages` Tables

Stores multi-turn grounded dialogue between the student and the AI career mentor.

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5. Backend API Specification (FastAPI v1)

5.1 Core Configuration & Security Framework

Located in `backend/app/core/`: - **`config.py`**: Manages environment variables using Pydantic `BaseSettings`. Handles `DATABASE_URL`, `JWT_SECRET_KEY`, `ALGORITHM` (HS256), `ACCESS_TOKEN_EXPIRE_MINUTES` (43,200 min / 30 days), and optional cloud LLM API keys. - **`security.py`**: - `verify_password(plain, hashed)`: Cryptographic verification via Bcrypt. - `get_password_hash(password)`: Generates salted Bcrypt password hash. - `create_access_token(data, expires_delta)`: Encodes JWT with expiration and user ID. - Dependency guards: `get_current_user` decodes token and queries database; `require_role(role)` enforces role-based authorization.

5.2 RESTful Domain Routers & Endpoints

Domain	Route Prefix	Key Endpoints	Auth Level	Purpose
Auth	`/api/v1/auth`	`POST /register` `POST /login` `GET /me` `POST /forgot-password` `POST /reset-password`	Public / Bearer	Registration, authentication, token issuance, and password recovery.
Students	`/api/v1/students`	`GET /me` `PATCH /me` `POST /skills` `DELETE /skills/{id}` `POST /projects`	Student	Profile reading, metadata editing, skill management, and project portfolio tracking.
Careers	`/api/v1/careers`	`GET /roles` `GET /analysis` `GET /skills`	Public / Student	Career role catalogue, skill gap analysis against target role, and master skill taxonomy.
Roadmaps	`/api/v1/roadmaps`	`GET /` `POST /generate` `PATCH /{id}/milestones/{mid}`	Student	Dynamic milestone generation, progress percentage calculation, and milestone completion toggling.
Jobs	`/api/v1/jobs`	`GET /` `GET /{id}` `GET /matched/list` `POST /{id}/save` `DELETE /{id}/save`	Public / Student	Opportunity exploration with category and workplace filtering, matched ranking, and bookmarking.
Applications	`/api/v1/applications`	`POST /submit` `GET /my-applications`	Student	Job application submission with cover letters and student application status tracking.

Domain	Route Prefix	Key Endpoints	Auth Level	Purpose

Recruiters	`/api/v1/recruiter`	`GET /dashboard`
		`GET /jobs`
		`POST /jobs`
		`PATCH /jobs/{id}`
		`GET /applications`
		`PATCH /applications/{id}/status`
		`GET /candidates`
			Recruiter	Recruiter telemetry, posting management, applicant pipeline status transitions, and talent search.

Resumes	`/api/v1/resumes`	`POST /analyze` `GET /history`	Student	Safe multipart PDF/TXT text extraction, ATS keyword matching, and impact verb enhancement.

Interviews	`/api/v1/interviews`	`POST /start`
		`POST /{id}/submit-answer`
		`GET /history`
			Student	AI mock interview session creation, multi-question rubric evaluation, and feedback scoring.

Assistant	`/api/v1/assistant`	`POST /chat`
		`GET /conversations`
		`GET /conversations/{id}`
			Student / User	Grounded career mentor dialogue referencing live platform jobs and student profile data.

Resources	`/api/v1/resources`	`GET /` `GET /role/{slug}`	Public / Student	Curated documentation, official tutorials, and open courses for skill progression.
Analytics	`/api/v1/analytics`	`GET /overview`	Public / All	Market demand frequency distribution, hiring growth trends, and observation telemetry.

6. Frontend Architecture & Client State

6.1 React 19 + TypeScript + Vite Architecture

- **Single Page Application (SPA)** architecture bundled with Vite 8.
- Zero reliance on heavyweight CSS frameworks; completely constructed with tailored **Vanilla CSS Design Tokens** (`index.css`), yielding ultra-low bundle sizes and instant load times.
- Fully typed data layer: interfaces in `src/api/` match backend Pydantic schemas.

6.2 Routing Hierarchy & Route Guards

Configured via `react-router-dom` v7 in `src/App.tsx`:

```
/                          → LandingPage (Master 3D WebGL Continuous Journey)
│
├── Public Layout (Navbar + Content + Footer)
│   ├── /about               → AboutPage
│   ├── /features            → FeaturesPage
│   ├── /opportunities       → PublicOpportunitiesPage
│   ├── /login               → LoginPage (Interactive with 1-Click Demo Personas)
│   ├── /register            → RegisterPage
│   ├── /forgot-password     → ForgotPasswordPage
│   ├── /reset-password      → ResetPasswordPage
│   ├── /privacy             → PrivacyPage
│   └── /terms               → TermsPage
│
├── Student Layout (Slim Top Navigation Bar, Guarded)
│   ├── /student/dashboard   → DashboardPage
│   ├── /student/profile     → ProfilePage
│   ├── /student/skills      → SkillsPage
│   ├── /student/career-guidance → CareerGuidancePage
│   ├── /student/skill-gap   → SkillGapPage
│   ├── /student/roadmap     → RoadmapPage
│   ├── /student/learning    → LearningPage
│   ├── /student/projects    → ProjectsPage
│   ├── /student/resume      → ResumePage
│   ├── /student/interview   → InterviewPage
│   ├── /student/opportunities→ OpportunitiesPage
│   ├── /student/applications → ApplicationsPage
│   ├── /student/saved       → SavedJobsPage
│   ├── /student/assistant   → AssistantPage
│   ├── /student/notifications→ NotificationsPage
│   └── /student/settings    → SettingsPage
│
└── Recruiter Layout (Slim Top Navigation Bar, Guarded)
    ├── /recruiter/dashboard  → RecruiterDashboardPage
    ├── /recruiter/company    → CompanyPage
    ├── /recruiter/jobs       → RecruiterJobsPage
    ├── /recruiter/jobs/new   → PostJobPage
    ├── /recruiter/applications → RecruiterApplicationsPage
    ├── /recruiter/candidates → CandidatesPage
    ├── /recruiter/analytics  → AnalyticsPage
    └── /recruiter/settings   → RecruiterSettingsPage
```

6.3 Global State Management (AuthContext & ToastContext)

1. `AuthContext` (`src/context/AuthContext.tsx`): - Manages active user authentication status, JWT token persistence in `localStorage` , and role switching. - Provides methods: `login(email, password)` ,

`register( ... )`, `logout()`, `updateUser( ... )`, and `switchPersona(role)`. - On application mount, verifies stored tokens against the backend `/auth/me` endpoint.

2. **`ToastContext` (`src/context/ToastContext.tsx`)**: - Manages queue-based floating feedback notifications (`success`, `error`, `info`, `warning`). - Renders animated toast popups with auto-dismiss timers and Lucide icon badges.

6.4 Resilient API Layer & Offline Demo Data Fallback

Located in `src/api/client.ts` and `src/api/demoData.ts`: - **Centralized HTTP Client**: Wraps native `fetch` with token injection, timeout controllers, and automatic JSON response parsing. - **Backend Health Probe**: The `checkHealth()` function monitors backend availability and latency, driving the live pulse indicator in the top navbar. - **Graceful Offline Degradation**: When operating disconnected from the backend, API client methods fallback transparently to rich demo data (`demoData.ts`), ensuring zero-downtime demonstration capability.

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7. Visual Design System & UI Architecture

7.1 Design Philosophy: Cinematic Cybernetic Glassmorphism

The visual design language communicates precision, technological sophistication, and academic rigor: - **Atmospheric Depth**: Deep void backgrounds paired with subtle radial glows and multi-layered backdrop blur. - **Frosted Glass Elevation**: Translucent card surfaces with high saturation, inset highlights, and gradient borders that simulate real frosted glass. - **Tactile Responsiveness**: Interactive elements provide instant visual feedback through micro-animations, scale shifts, and glowing borders.

7.2 Design Tokens, Colors & CSS Variables

Extracted from `landing page/src/index.css`:

```
:root {
  /* Surfaces & Backgrounds */
  --bg-primary: #05070D;              /* Deep Cyber Void */
  --bg-secondary: #0A0E18;            /* Dark Midnight Navy */
  --bg-card: rgba(10, 15, 29, 0.60);  /* Translucent Glass Base */
  --bg-card-hover: rgba(14, 21, 40, 0.72);
  --bg-glass: rgba(255, 255, 255, 0.035);
  --bg-glass-active: rgba(255, 255, 255, 0.08);

  /* Typography Colors */
  --text-primary: #F5F7FF;            /* Pure Crisp White */
  --text-secondary: #8D96AA;          /* Balanced Neutral Slate */
  --text-muted: #5A6478;              /* Subtle Dim Gray */

  /* Borders & Glass Dividers */
  --border-subtle: rgba(255, 255, 255, 0.09);
  --border-focus: rgba(255, 255, 255, 0.20);
  --border-highlight: rgba(167, 139, 250, 0.4);
  --border-glass: rgba(255, 255, 255, 0.12);

  /* Curated Vivid Accents */
  --accent-lavender: #A78BFA;         /* Primary Brand Glow */
  --accent-violet: #8B5CF6;           /* Deep Electric Violet */
  --accent-cyan: #38BDF8;             /* Algorithmic Sky Cyan */
  --accent-mint: #34D399;             /* Success & Verification Emerald */
  --accent-amber: #FBBF24;            /* Caution & Alert Gold */

  /* Soft Ambient Glows */
  --glow-lavender: rgba(167, 139, 250, 0.15);
  --glow-cyan: rgba(56, 189, 248, 0.15);
  --glow-mint: rgba(52, 211, 153, 0.12);

  /* Border Radii */
  --radius-sm: 8px;
  --radius-md: 12px;
  --radius-lg: 16px;
  --radius-xl: 24px;
  --radius-full: 9999px;

  /* Cubic Bezier Physics */
  --transition-smooth: all 0.25s cubic-bezier(0.16, 1, 0.3, 1);
  --transition-spring: all 0.4s cubic-bezier(0.34, 1.56, 0.64, 1);
}
```

7.3 Typography Hierarchy & Font Pairing

- **Display Headings (`font-family: 'Space Grotesk'`)**: Applied to all `h1` through `h6` elements. Features geometric, modern letterforms with negative tracking (`letter-spacing: -0.025em`) for a futuristic, cinematic feel.
- **Body & Data Typography (`font-family: 'Inter'`)**: Applied to body copy, table cells, metric labels, and form controls. Ensures optimal legibility across high-density technical dashboards.

7.4 Glassmorphic Panels & Surface Elevation

The `.glass-panel` utility class implements multi-layered depth:

```
.glass-panel {
  background: linear-gradient(135deg, rgba(255, 255, 255, 0.05) 0%, rgba(255, 255, 255, 0.01) 100%), var(-
  backdrop-filter: blur(24px) saturate(180%);
  -webkit-backdrop-filter: blur(24px) saturate(180%);
  border: 1px solid var(--border-glass);
  border-radius: var(--radius-lg);
  box-shadow: 0 20px 48px -12px rgba(0, 0, 0, 0.5),
              inset 0 1px 0 rgba(255, 255, 255, 0.14),
              inset 0 -1px 0 rgba(0, 0, 0, 0.3);
  transition: var(--transition-smooth);
}
```

7.5 Sleek Top Navigation Bar Redesign

In Phase 9, traditional bulky 260px vertical sidebars were replaced by a modern, high-performance horizontal **Top Navigation Bar** (`height: 54px` / `66px`): - Pushes navigation links to the horizontal header, freeing up 100% of horizontal viewport width for wide data tables, radar charts, and candidate pipelines. - Features a live **AI Engine pulse indicator** showing backend status and latency. - Profile pill displaying student initials, quick notification badges, and dropdown menus for workspace settings and sign out.

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8. 3D WebGL Journey Canvas (Three.js Engine)

8.1 WebGL Pipeline & Continuous Spline Path

Located in `landing page/src/three/JourneyCanvas.tsx`: - Rendered via a hardware-accelerated `THREE.WebGLRenderer` with `antialias: true`, `alpha: true`, and `ACESFilmicToneMapping`. - Uses a `THREE.CatmullRomCurve3` centripetal spline with 12 mathematical control points traversing from the Hero origin ($Z = +6.0$) to the Future Horizon ($Z = -75.0$):

```
(1.4, -0.95, 6)   ──> Hero Origin (Traveler at Start)
(1.2, -0.9, 2.5)  ──> Curved Hero Transition
(0.5, -0.6, -2)   ──> Transition to Career Network
(0.0, 0.4, -13)   ──> Core Sphere (Career Intelligence Orbit)
(1.2, -0.8, -38)  ──> Learning Roadmap (Ascending Steps)
(0.4, -0.2, -55)  ──> Opportunity Matcher (Network Graph)
(0.0, 0.4, -75)   ──> Future Horizon (Summit & Archway)
```

8.2 Scroll-Synchronized Camera Choreography

- As the user scrolls through the landing page, `scrollProgress` (normalized 0.0 to 1.0) is tracked via a passive window listener.
- The camera position and look-at target interpolate smoothly along the 3D curve using linear easing (`MathUtils.lerp`).
- Damped mouse parallax adds subtle organic movement as the cursor moves across the viewport.

8.3 Interactive Visual Objects & Chapter Themes

1. **Spline Tube Geometry**: Built using `THREE.TubeGeometry` with 180 segments. Vertex colors blend seamlessly from Lavender (`#A78BFA`) through Sky Cyan (`#38BDF8`) to Mint (`#34D399`).
2. **Photon Traveler Bead**: A glowing white mesh accompanied by a `PointLight` that traverses the tube path in sync with scroll progress.
3. **Hero Traveler Silhouette**: Minimalist geometric humanoid figure on a glowing ring pedestal overlooking the journey ahead.
4. **Core Sphere Group**: A faceted icosahedron surrounded by an animated wireframe outer shell and orbital particle rings, symbolizing career intelligence.
5. **Roadmap Milestone Markers**: Ascending crystalline platforms representing educational milestones.
6. **Opportunity Network Nodes**: Interconnected cluster of glowing vertices representing matched industry roles.
7. **Future Horizon Summit**: Triumphant portal archway and summit traveler silhouette surrounded by radiant ambient particles.

8.4 Performance & Accessibility Adaptations

- Detects the `prefers-reduced-motion` media query to disable heavy parallax animations for users with motion sensitivity.
- Pixel ratio is clamped to $\min(\text{devicePixelRatio}, 2)$ to maintain 60 FPS on high-DPI retina displays.
- Efficiently disposes geometries, materials, and requestAnimationFrame loops on component unmount.

9. Comprehensive Page-by-Page UI/UX Specifications

9.1 The Master 3D Landing Page (6 Chapters)

Chapter 1: Hero Section (`Hero.tsx`)

- High-impact headline: *"Bridge Your Skills. Build Your Tomorrow."*
- Interactive primary CTAs: *"Explore Career Intelligence"* and *"For Tech Recruiters"*.
- Floating live metrics banner displaying platform stats (94% placement rate, 4 curated specializations, 100+ active roles).
- Smooth entry point to the 3D journey path.

Chapter 2: Career Network (`CareerNetwork.tsx`)

- Visual graph showing the interconnected relationship between academic curricula and industry role requirements.
- Highlights the flow of student competencies into machine learning, data engineering, full-stack development, and DevOps tracks.

Chapter 3: Career Intelligence (`CareerIntelligence.tsx`)

- Interactive demonstration of the **6-axis Skill Gap Radar**.
- Live role switcher allowing visitors to preview skill gaps for AI Engineer vs. Full-Stack Systems Engineer vs. Data Scientist roles.
- Explains the deterministic scoring methodology with clear visual formulas.

Chapter 4: Dynamic Learning Roadmap (`LearningRoadmap.tsx`)

- Step-by-step milestone timeline showcasing estimated weekly hours, deliverables, and acquired competencies.
- Interactive milestone toggle previewing real-time progress percentage updates.

Chapter 5: Opportunity Matcher (`OpportunityMatcher.tsx`)

- Connected directly to the live backend `/api/v1/jobs` API.
- Displays realistic job cards with transparent match score breakdowns (e.g., *"92% Match: 5/5 Required Skills, +5% Project Bonus"*).
- Category filters (AI/ML, Web Systems, Cloud) and workplace badges (Remote, Hybrid, Onsite).

Chapter 6: Final Call to Action (`FinalCTA.tsx`)

- High-energy summit conclusion with celebratory particle effects.
- Dual action buttons directing students and recruiters to instantaneous registration.

9.2 Public & Authentication Pages

Interactive Sign In Page (`LoginPage.tsx`)

Redesigned in Phase 10 with interactive usability features: - **1-Click Demo Persona Cards**: - **Alex Rivera (Student)**: Auto-fills `student@careerbridge.ai` / `student123` with a single click. - **Sarah Chen (Recruiter)**:

Auto-fills `recruiter@deepmind.com` / `recruiter123` with a single click. - **Interactive Password Controls**: `Eye` / `EyeOff` visibility toggle buttons. - **Real-Time Caps Lock Detection**: Warning banner appears instantly when Caps Lock is active. - **Dynamic Email Validation**: Real-time syntax validation with green checkmark indicators. - **Stepped Authentication Progress**: Displays multi-step feedback (*"Verifying credentials..."* → *"Retrieving JWT token..."* → *"Redirecting..."*) followed by celebratory Canvas Confetti on success.

Registration Page (`RegisterPage.tsx`)

- Role selector toggle (`Student` vs. `Recruiter`).
- Full form validation with password strength meter, email uniqueness check, and terms acceptance.

Additional Public Pages

- **About Page (`AboutPage.tsx`)**: Platform mission, DeepMind-inspired algorithmic ethics, and founding story.
- **Features Page (`FeaturesPage.tsx`)**: In-depth breakdown of all 6 platform intelligence engines.
- **Public Opportunities (`PublicOpportunitiesPage.tsx`)**: Searchable and filterable job directory accessible to unregistered visitors.
- **Legal Pages (`PrivacyPage.tsx` , `TermsPage.tsx`)**: Complete GDPR-compliant data privacy policies and terms of service.

9.3 Student Portal (16 Sub-Pages)

1. **Dashboard (`DashboardPage.tsx`)**: Central command center featuring the career readiness gauge, active milestone countdown, top matched opportunities, and quick action cards.
2. **Profile (`ProfilePage.tsx`)**: Academic details, target career role, biographical summary, and repository links (GitHub, LinkedIn, Portfolio).
3. **Skills (`SkillsPage.tsx`)**: Comprehensive skill inventory management. Add skills with proficiency sliders (1–5) and provide verifiable GitHub project evidence.
4. **Career Guidance (`CareerGuidancePage.tsx`)**: High-level guidance overview comparing candidate skills against industry career paths.
5. **Skill Gap Analysis (`SkillGapPage.tsx`)**: Interactive 6-axis SVG radar visualization, critical missing competencies list, and tailored recommendations.
6. **Dynamic Roadmap (`RoadmapPage.tsx`)**: Complete multi-week curriculum broken into sequential milestones. Check off deliverables to trigger real-time progress recalculation.
7. **Curated Learning (`LearningPage.tsx`)**: Filterable library of free and open-access courses, documentation, and tutorials mapped directly to student skill gaps.
8. **Projects Portfolio (`ProjectsPage.tsx`)**: Showcase engineering projects with repository URLs, live demo links, and tagged technology stacks that feed the $+5\%$ match bonus.
9. **AI Resume Diagnostics (`ResumePage.tsx`)**: PDF/TXT drag-and-drop uploader, ATS readiness score gauge, missing keyword alerts, and high-impact action verb suggestions.
10. **AI Mock Interview (`InterviewPage.tsx`)**: Interactive practice simulator with question timer, answer input, 4-axis rubric scoring, and model benchmark comparisons.
11. **Opportunity Explorer (`OpportunitiesPage.tsx`)**: Full job search interface with transparent match percentages, requirement breakdowns, and one-click application submission.
12. **Applications Tracker (`ApplicationsPage.tsx`)**: Real-time status pipeline tracking submitted applications (`SUBMITTED` → `UNDER_REVIEW` → `SHORTLISTED` → `OFFERED`).

13. **Saved Jobs (`SavedJobsPage.tsx`)**: Bookmarked positions saved for later review.
14. **AI Assistant (`AssistantPage.tsx`)**: Grounded multi-turn conversational mentor answering career questions using live platform database records.
15. **Notifications (`NotificationsPage.tsx`)**: System alerts regarding application status changes, new matching jobs, and milestone reminders.
16. **Settings (`SettingsPage.tsx`)**: Password changes, notification preferences, and account management.

9.4 Recruiter Portal (8 Sub-Pages)

1. **Recruiter Dashboard (`RecruiterDashboardPage.tsx`)**: Key recruiting metrics (active job postings, total applicants, shortlisted talent, pipeline velocity).
2. **Company Profile (`CompanyPage.tsx`)**: Manage company branding, culture description, website link, and industry tags.
3. **Job Postings Management (`RecruiterJobsPage.tsx`)**: List and status view of all open, paused, and closed company listings.
4. **Post New Role (`PostJobPage.tsx`)**: Structured job creation wizard with required vs. preferred skill tag pickers, salary ranges, and workplace type selection.
5. **Applicant Pipeline (`RecruiterApplicationsPage.tsx`)**: Pipeline management interface for moving candidates across stages (`UNDER_REVIEW`, `SHORTLISTED`, `INTERVIEW`, `OFFERED`, `REJECTED`).
6. **Candidate Discovery (`CandidatesPage.tsx`)**: Search talent database by skill keywords, match readiness, and academic degree, with privacy-guarded outreach.
7. **Talent Analytics (`AnalyticsPage.tsx`)**: Labor market demand heatmaps, skill frequency distributions, and candidate supply curves.
8. **Recruiter Settings (`RecruiterSettingsPage.tsx`)**: Hiring team preferences, email alerts, and organizational settings.

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10. Testing, Security & Deployment Guide

10.1 Backend Automated Test Suite (Pytest)

Located in `backend/tests/`: - **17 automated test cases** validating core platform functionality: 1. `test_auth.py`: Registration, duplicate email rejection, Bcrypt verification, JWT token issuance, and `/me` validation. 2. `test_career_intelligence.py`: Deterministic skill gap scoring formulas, verification bonus factor calculation, and radar axis projection. 3. `test_roadmap.py`: Roadmap milestone generation and completion percentage recalculation. 4. `test_jobs_applications.py`: 70/30 matching mathematics, project bonus application, application lifecycle transitions, and duplicate prevention. 5. `test_ai_services.py`: Resume keyword extraction, interview rubric scoring, and graceful AI fallback. - **Execution Command**: `bash cd backend python -m pytest tests -v` - **Result**: **17 passed in ~1.6s**.

10.2 Frontend Build Verification

- Validated via TypeScript compiler and Vite production build: `bash cd "landing page" npm run build`
- **Result**: `tsc -b && vite build` succeeds with **0 errors in <500ms**, generating an optimized production bundle.

10.3 Security Auditing & Privacy Boundaries

- **Password Hashing**: Bcrypt with auto-generated salt; plain text passwords are never logged or stored.
- **Token Security**: HS256 signed JWT tokens containing user ID subject claims.
- **Privacy Guards**: Student email and phone numbers are hidden from recruiter discovery searches until an application is officially moved to `SHORTLISTED` or `INTERVIEW` stage.
- **SQL Injection Prevention**: SQLAlchemy parameterized queries eliminate SQL injection vulnerabilities.
- **XSS Mitigation**: React JSX automatic string escaping prevents cross-site scripting attacks.

10.4 Local & Production Deployment Setup

One-Click Local Launch (Windows)

1. Run `start_backend.bat` in one terminal window (starts Uvicorn on port 8000).
2. Run `start_frontend.bat` in a second terminal window (starts Vite on port 5173).
3. Access the application at `http://localhost:5173`.

Production Deployment Checklist

1. Configure a production PostgreSQL database instance.
2. Set environment variables: `env DATABASE_URL=postgresql+psycopg2://user:password@host:5432/careerbridge JWT_SECRET_KEY=generate_a_cryptographically_secure_random_string_64_chars ENVIRONMENT=production`
3. Run database migrations: `alembic upgrade head`
4. Seed initial catalogue: `python -m app.db.seed`
5. Build the React frontend: `npm run build` in `landing page/` and serve static assets via Nginx or Cloudflare Pages.

6. Run Uvicorn behind a production reverse proxy with Gunicorn worker processes: `bash uvicorn app.main:app --host 0.0.0.0 --port 8000 --workers 4`

---

11. Project Conclusion & Summary Table

CareerBridge AI represents a complete, mathematically grounded, and visually stunning career platform engineered for modern technical talent.

Dimension	Implementation Details
Frontend Framework	React 19.2 + TypeScript 5.8 + Vite 8.3 + React Router DOM 7
Visual Styling	Bespoke Dark Glassmorphism tokens (`index.css`), Space Grotesk + Inter
3D Visualization	Three.js 0.174 continuous WebGL Catmull-Rom spline with shader gradients
Backend Framework	Python 3.13 + FastAPI 0.115 + Pydantic v2
Database & ORM	SQLAlchemy 2.0 + SQLite (Default Zero-Setup) / PostgreSQL (Production)
Schema Migrations	Alembic 1.20 with SQLite batch migration mode
Security & Auth	Salted Bcrypt + HS256 JWT tokens + RBAC Dependency Guards
AI Intelligence	Local Intelligent Deterministic Engines + Pluggable Cloud LLM Fallback
Automated Testing	17 Pytest automated test cases passing in 1.6s
Production Build	`npm run build` passing with 0 TypeScript/bundler errors

Developed for CareerBridge AI — AI-Powered Career Intelligence & Recruitment Platform.